Underground line Methodology

A methodology to design an underground line

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Abstract

This document presents a structured methodology for the sizing and calculation of high-voltage (HV) underground cables used to connect electrical substations to the point of interconnection (POI). The methodology addresses the technical parameters that influence cable selection, including current-carrying capacity, voltage drop, and short-circuit withstand capability. It integrates industry standards and best practices to ensure compliance and operational efficiency.

By considering environmental conditions, trench installation method, and system load requirements, this approach provides a reliable framework for designing HV cable systems that meet both present and future grid demands.

The main topics that will be covered in this document to explain the design of the HV underground line are the following:

- The cable sizing
- The equipment selection
- The trench sizing
- Electrical parameters calculation

Note: All the calculations that are presented in this document are carried out according to the IEC standards.

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Chapter 1

Underground line criteria

In this chapter the HV underground line's main elements and design criteria will be described.

This chapter outlines the key parameters and design criteria that influence the selection and configuration of the HV underground line, focusing on the voltage level and circuits' formation that directly impact trenching, cable laying, and thermal performance.

1.1 Design criteria

The underground line under consideration in this document has a transmission nature, designed to connect the substation to the point of interconnection (POI) or the intermediate substation to the pooling substation. HV underground lines are typically used in areas where overhead lines are impractical due to environmental, urban, or regulatory constraints.

Key characteristics of the HV transmission line include:

- Voltage level: minimum voltage of 46kV and maximum voltage of 550kV.
- Cable type: XLPE-insulated, single-core, maximum of 3 circuits.
- Formation: flat or trefoil formation.
- Installation method: Cable trenches
- Compliance standards: IEC standards

In this methodology, several electrical requirements in designing an underground line were considered, these considerations are defined to guarantee the stability and the capability of the line.

1.2 Mechanical considerations

Mechanical considerations play a vital role in the design and installation of HV underground cables. The physical arrangement of the cables and the method of installation directly influence the system's thermal behavior and long-term reliability. This section outlines the key mechanical aspects considered in the design.

1.2.1 Installation type

The selected installation method for the HV cable is direct burial in trench, which is a common and cost-effective approach for underground transmission lines. This method involves placing the cables directly into a prepared trench, typically with layers of bedding and backfill materials to provide mechanical protection and thermal stability.

Key considerations for direct burial include:

- Trench dimensions: Determined based on cable size, number of circuits, formation type, and required separation distances. The minimum trench width is set to be 0.4m.
- Bedding and backfill consideration: Use of thermally stable materials (e.g., sand or thermal backfill) to ensure consistent heat dissipation.
- Mechanical protection: Warning tapes, concrete tiles, or protective covers may be used to prevent accidental damage during future excavation.
- Depth of burial: Ranges from 0.5 to 2 meters.

Direct burial is preferred for its simplicity and lower cost, but it requires careful planning to ensure long-term reliability and ease of maintenance.

1.2.2 Formation type

The cable formation refers to the geometric arrangement of the three phases of the HV cable. In this methodology, the two most common configurations, trefoil and flat are used:

Trefoil Formation:

Cables are laid in a triangular (cloverleaf) configuration as shown in figure 1.1. Minimizes the loop area, reducing electromagnetic interference (EMI) and induced voltages in nearby metallic structures. This formation offers better short-circuit performance due to reduced magnetic forces. Furthermore, it is typically used in areas with limited trench width or where EMI mitigation is critical.

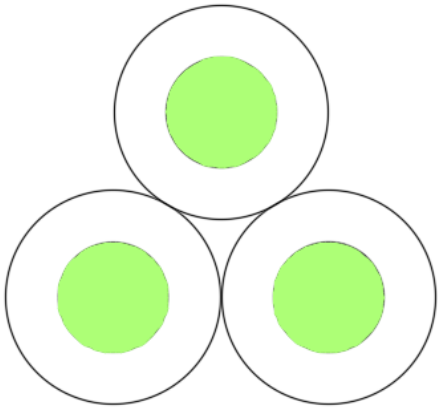


Figure 1.1: Trefoil formation illustration

Flat Formation:

Cables are laid side-by-side in a horizontal line, as shown in figure 1.2. Easier to install and inspect, especially in wide trenches. Higher mutual inductance and magnetic field exposure compared to trefoil. May require greater spacing or shielding to manage thermal and electromagnetic effects. In this design, the distance between the phases is considered zero.

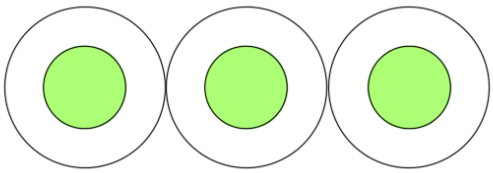


Figure 1.2: Flat formation illustration

The choice between trefoil and flat formation depends on space availability, thermal performance, electromagnetic compatibility, and installation constraints. However, in this methodology, it is a user's preference.

1.3 Type of terminals

Cable terminals are used to connect the HV underground cable to equipment at the substation or the point of interconnection. These components must safely transition the cable from its shielded, insulated form to an open-air or enclosed connection point.

Key Requirements:

- Voltage rating: Terminals are rated for the system voltage.
- Compatibility: The terminals are considered to match the cable's insulation, conductor, and screen design.
- Type of terminal: Only outdoor terminals are considered in this methodology.

Installation Location:

The terminals are typically installed at the substation's both ends and/or at POI. In this design, the terminals are considered mounted on structure or the gate of the substation.

1.4 Type of joints

Cable joints are essential components in HV underground systems, used to connect cable segments along the route. In this methodology we are considering joints in every 400 meters of the cable.

Selection Criteria:

- Voltage level compatibility: Joints are rated for the system's operating voltage.
- Cable type matching: Joints are considered compatible with the insulation type, conductor material, and screen/sheath configuration.

- Thermal and electrical performance: Joints are selected so that they maintain the same current-carrying capacity and thermal characteristics as the cable itself.

Installation Considerations:

The joints in this methodology are considered installed in accessible joint pits or vaults for inspection and maintenance.

Chapter 2

Conductor sizing

Proper cable sizing is crucial in underground line design to ensure electrical safety, system reliability, and operational efficiency. Accurate sizing considers factors such as current-carrying capacity, voltage drop, and short circuit current, ensuring the system performs optimally under all expected conditions. Ultimately, correct cable sizing supports long-term durability and minimizes maintenance and operational risks.

2.1 Conductor's type

In the sizing of high-voltage underground cables, the choice of conductor material plays a critical role in determining the cable's electrical and thermal performance. For this methodology, XLPE (cross-linked polyethylene) insulated cables are selected due to their excellent thermal stability and high dielectric strength. The conductors used are either aluminum or copper; copper offers superior conductivity and is preferred in installations requiring compact sizing and high current capacity, while aluminum provides a cost-effective alternative with lower weight, making it suitable for longer runs.

All cable design and sizing are performed in accordance with the IEC 60287 [1] and IEC 60228 [2] standards, which provide a comprehensive methodology for determining the continuous current-carrying capacity of cables under various installation conditions. Furthermore, all the cables selected for the sizing are chosen from the manufacturer ABB [3] ranging from 95mm² to 2000mm².

2.2 Sizing criteria

The sizing of the underground cable is determined through an optimization process aimed at finding the smallest conductor capable of withstanding the operational conditions of the line. Specifically, the selected conductor must be able to carry the operating current, endure short-circuit conditions, and ensure that the voltage drop does not exceed a user-defined threshold.

2.2.1 Current capacity

The maximum current a conductor can withstand is determined by its configuration and the operating conditions. Under standard conditions, the base current I_s that the conductor can withstand depends on where the cable is laid, its formation, the bonding and the operating temperature.

The current carrying capacity criterion is a fundamental aspect of cable sizing, ensuring that the conductor can work under the expected load current without exceeding its admissible temperature limits. For the calculation of the admissible current, manufacturer's datasheets will been used, and different safety factors will be applied to the admissible current depending on some installation parameters.

The following correction factors are applied to the HV cables:

- **Laying depth factor**: The depth at which cables are buried affects their heat dissipation. Cables installed deeper in the ground experience reduced cooling, resulting in a lower admissible current.
- **Ground temperature factor**: The thermal environment of the soil influences current capacity. For ground temperatures below 20ºC, the admissible current increases, while temperatures above 20ºC reduce it.
- **Thermal resistivity factor**: Lower resistivity improves heat dissipation and increases current capacity, whereas higher resistivity has the opposite effect.
- **Distance between circuits and groups of circuits**: The number of circuits and the spacing between them significantly impact thermal performance. Closely spaced or grouped circuits generate more heat, reducing the admissible current due to limited heat dissipation.

For this methodology, the following assumptions have been made when sizing and rating the cables for the standard conditions:

- Three-phase group of single-core cables.
- The soil temperature equals 25ºC if no information is available.
- Laying depth of 1.0m.
- Distance between cable axes in flat formation of $70mm + De$, where De is the exterior diameter of the conductor.
- The soil resistivity is 1 $K \cdot m/W$ if no information is available.
- Cables are always directly buried.
- Operating temperature of 90ºC.
- For cables carrying between 10 to 70kV, 35mm^2 of copper screen, and 95mm^2 for higher voltages.

When conditions are not standard, several factors d_i are considered in the form of corrective coefficients or derating factors. In particular, for this methodology, the factors considered are the following: laying depth, ground temperature, ground thermal resistivity and the number of groups of cables on the ground. These factors are based on the IEC 60287 standard [1], showed in Table 2.1, Table 2.2, Table 2.3 and Table 2.4.

Laying depth, m	Rating factor
0.5	1.1
0.7	1.05
0.9	1.01
1	1
1.2	0.98
1.5	0.95

Table 2.1: Rating factor for laying depth

Conductor temperature	Ground temperature, °C							
°C	**10**	**15**	**20**	**25**	**30**	**35**	**40**	**45**
90	1.07	1.04	1	0.96	0.93	0.89	0.84	0.8
65	1.11	1.05	1	0.94	0.88	0.82	0.74	0.66

Table 2.2: Rating factor for ground temperature based on conductor temperature

Thermal resistivity, K·m/W	Rating factor
0.7	1.14
1	1
1.2	0.93
1.5	0.84
2	0.74
2.5	0.67
3	0.61

Table 2.3: Rating factor for ground thermal resistivity

Distance cc between groups (mm)	1	2	3	4	5	6	7	8	9
100	1.00	0.76	0.67	0.59	0.55	0.51	0.49	0.47	0.46
200	1.00	0.81	0.71	0.65	0.61	0.49	0.56	0.53	0.52
400	1.00	0.85	0.77	0.72	0.69	0.66	0.64	0.63	0.62
600	1.00	0.88	0.81	0.77	0.74	0.72	0.71	0.70	0.69
800	1.00	0.89	0.84	0.81	0.79	0.77	0.76	0.76	0.75
2000	1.00	0.96	0.93	0.92	0.91	0.91	0.91	0.90	0.90

Table 2.4: Rating factor for groups of cables in the ground

In light of the above, the current cable capacity is computed by multiplying the ampacity in standard conditions by the aforementioned derating factors as per Equation 2.1.

$$I_{ccc} = I_s \cdot \prod d_i \tag{2.1}$$

Where:

- I_{ccc} is the current cable capacity $[A]$
- I_s is the current capacity under standard conditions $[A]$
- d_i is the i-th derating factor in a range between 0 and 1

2.2.2 Short Circuit Current

When a short-circuit occurs, the amount of current flowing through the conductor might surpass the nominal current during short periods of time, heating up the conductor. It is necessary to verify that the proposed cross section can withstand the maximum short circuit current. The smallest cross section capable of withstanding the short circuit current is computed as per Equation 2.2.

$$S = \frac{I_{sc} \cdot \sqrt{t}}{K \cdot \sqrt{\ln\left(\frac{\theta_f + \beta}{\theta_i + \beta}\right)}} \tag{2.2}$$

Where:

- S is the minimum required conductor cross-sectional area$[mm^2]$.
- I_{sc} is the short-circuit current (RMS) $[A]$, determined based on the voltage level:
  - 45kV to 123kV: 25000 A
  - 124kV to 200kV: 31500 A
  - 201kV to 420kV: 50000 A
  - Above 421kV: 63000 A
- t is the fault duration $[s]$; a standard value of 1 second is used
- K is a material-dependent constant (226 for copper, 148 for aluminum) $[As^2/mm^2]$
- θ_f is the final permissible conductor temperature $[C]$, set to 250C
- θ_i is the initial conductor temperature $[C]$, assumed to be 90C
- β is the reciprocal of the temperature coefficient of electrical resistance at 0C in $[K]$ (234.5 for copper and 228 for aluminum)

2.2.3 Voltage drop

A key criterion for sizing the underground line is the voltage drop. The cable needs to be sized ensuring that the voltage drop remains below a certain threshold defined by the user. This can be achieved by adjusting factors like the cross section, the impedance, number of circuits, and/or the length of the cable. A larger conductor or multiple parallel cables reduce the resistance and reactance, minimizing the voltage drop and ensuring efficient power transmission. The voltage drop is defined according to Equation 2.3.

$$Vd_{3ph}(\%) = \frac{\sqrt{3} \cdot I_b \cdot L \cdot [R \cdot cos\theta + X \cdot sin\theta]}{Vd_{3ph} \cdot N} \cdot 100 \tag{2.3}$$

Where:

- $Vd_{3ph}(\%)$ is the percentage voltage drop in a three-phase system
- I_b is the load current
- L is the length of the circuit $[km]$
- R is the resistance of the cable per unit length $[\Omega/km]$
- X is the reactance of the cable per unit length $[\Omega/km]$
- θ is the power factor angle $[rad]$
- Vd_{3ph} is the operating voltage on the underground cable $[V]$
- N is the number of circuits

2.3 Electrical Parameters

The electrical model presents an electrical circuit with different components that are used to calculate the voltage and current at the receiving end of the underground line.

2.3.1 Resistance

The resistance of the underground line is given by the manufacturer for a DC system under a specific operating temperature. This value must be converted to the applicable case of an AC system at a certain temperature of operation.

To compute the AC resistance of the cable, Equation (2.4) will be used.

$$R_{ac\theta} = R_{dc\theta}\left(1 + y_s + y_p\right) \tag{2.4}$$

Where:

- $R_{ac\theta}$ is the resistance of the cable AC at operating temperature [Ω/m]
- $R_{dc\theta}$ is the resistance of the cable DC at operating temperature [Ω/m] calculated according to Equation (2.5)
- y_s is the skin effect calculated according to Equation (2.6)
- y_p is the proximity effect calculated according to Equation (2.8)

The calculation of the DC resistance at the operating temperature is as follows.

$$R_{dc\theta} = R_{dc20C}\left(1 + \alpha_c\left(\theta_c - 20\right)\right) \tag{2.5}$$

Where:

- $R_{dc\theta}$ is the resistance of the cable DC at operating temperature [Ω/m]
- R_{dc20C} is the resistance of the cable DC at 20°C. This value is provided in the manufacturer´s sheet [Ω/m]
- α_c is the temperature coefficient of the conductor's material per K at 20°C. Values are taken from IEC 60287-1-1 ($\alpha_c = 0.00393$ for copper and $\alpha_c = 0.00403$ for aluminum)
- θ_c is the operating temperature of the cable. This value is equal to 90 ° C for XLPE insulated cables, which will be the only insulator used to size the cables in this methodology.

The skin effect represents the tendency of the electrons to flow through the exterior perimeter of the cable. This means that the current is not equally distributed across the cable's section, increasing the effective resistance of the cable. Equation (2.6) shows the calculation of the skin effect for the cables:

$$y_s = \frac{x_s^4}{192 + 0.8 \cdot x_s^4} \tag{2.6}$$

Where:

- y_s is the skin effect of the cable
- x_s is a constant calculated according to Equation (2.7)

$$x_s^4 = \left(\frac{8\pi f}{R_{dc\theta}} k_s \cdot 10^{-7}\right)^2 \tag{2.7}$$

Where:

- f is the frequency [Hz]
- $R_{dc\theta}$ is the DC resistance at operating temperature from Equation (2.5) [Ω/m]
- k_s is a constant given by the IEC in Figure 2.1. Round and stranded conductors will be considered, thus a value of 1 will be taken.

Type of Conductor	Dried and Impregnated?	k_s	k_p
Copper			
Round, stranded	Yes	1	0.8
Round, stranded	No	1	1
Round, segmental	-	0.435	0.37
Sector-shaped	Yes	1	0.8
Sector-shaped	No	1	1
Aluminium			
Round, stranded	Either	1	1
Round, 4 segment	Either	0.28	0.37
Round, 5 segment	Either	0.19	0.37
Round, 6 segment	Either	0.12	0.37

Figure 2.1: Constants k_s and k_p for skin and proximity effects calculation. Source: IEC 60287-1

The proximity effect on the other side occurs due to the proximity of other conductors generating magnetic fields, which affect the characteristics of the cable. This results in an increase in the resistance of the cable depending on the frequency of the line. The magnitude of the proximity effect is calculated using Equation (2.8).

$$y_p = \frac{x_p^4}{192 + 0.8 \cdot x_p^4} \left(\frac{d_c}{s}\right)^2 \left[0.312 \left(\frac{d_c}{s}\right)^2 + \frac{1.18}{\frac{x_p^4}{192+0.8 \cdot x_p^4} + 0.27}\right] \tag{2.8}$$

Where:

- y_p is the proximity effect
- x_p is a constant calculated according to (2.9)

- d_c is the diameter of the conductor [mm]
- s is the distance between conductor axes [mm]. This distance depends on how the cables are laid. For trefoil formation, $s = 4r$, where r is the radius of the conductor in mm. This is assuming that the insulator thickness is equal to the radius of the conductor. For flat formation, $s = D_e + 70mm$ is assumed, where D_e is the diameter of the conductor.

$$x_p^4 = \left(\frac{8\pi f}{R_{dc\theta}} k_p \cdot 10^{-7}\right)^2 \tag{2.9}$$

Where:

- f is the frequency of the line [Hz]
- $R_{dc\theta}$ is the DC resistance at operating temperature from Equation (2.5) [Ω/m]
- k_p is a constant given by the IEC in Figure 2.1. Round and stranded conductors will be considered, thus a value of 1 will be taken.

2.3.2 Reactance

A current-carrying conductor generates a magnetic field around itself. In AC systems, the time-varying nature of the current causes a changing magnetic field, which gives rise to inductance. When multiple conductors are present in an electrical configuration, the magnetic field produced by one conductor influences the others. This mutual interaction introduces a geometric dependency, meaning the physical arrangement of the conductors significantly affects the overall inductance of the transmission line.

To obtain the reactance of the cable, Equation (2.10) can be used:

$$X_L = 2\pi f L \tag{2.10}$$

Where:

- X_L is the inductive reactance of the cable [Ω/m]
- f is the frequency of the line [Hz]
- L is the inductance of the cable given by the manufacturer [H/m]. This value will vary depending on the formation (trefoil or flat) of the cables.

2.3.3 Capacitive Reactance and Susceptance

The capacitive reactance can be calculated using Equation (2.11)

$$X_c = \frac{1}{2\pi f C} \tag{2.11}$$

Where:

- X_c is the capacitive reactance of the cable [Ω/m]
- f is the line frequency [Hz]
- C is the capacitance of the cable given by the manufacturer [F/m]

The susceptance can be then calculated as the inverse of the reactance from Equation (2.12)

$$B = \frac{1}{X_c} = 2\pi f C \tag{2.12}$$

Where B is the susceptance of the line in [Ω^{-1}/ m^{-1}]

2.3.4 Zero sequence impedance

To calculate phase-to-ground short-circuit currents, it is essential to determine the zero-sequence impedance. This calculation follows the IEC 60909-3 standard [4], which is also applied to medium-voltage cables as described in the Electrical Methodology [5].

The first step in determining the zero-sequence impedance involves computing the soil penetration depth, denoted as δ, using Equation (2.13)

$$\delta = \frac{1.851}{\sqrt{2\pi f \frac{\mu_0}{\rho}}} \tag{2.13}$$

Where:

- δ is the equivalent soil penetration depth in [m].
- f is the system frequency in [Hz].
- μ_0 is the permeability of free space, which is equal to $4\pi \cdot 10^{-7}$ in [H/m]
- ρ is the soil resistivity in [Ω/ m]. For the purposes of this analysis, a soil resistivity of 300 Ω/m was assumed, which corresponds to a medium resistivity level typically found in geological formations such as limestone, sandstone, and shale.

The impedance can then be calculated using the following expression:

$$Z_0 = R_0 + jX_0 = \frac{R_1}{n} + \frac{3}{4}\pi f \mu_0 + jf\mu_0 \left(3 \cdot \ln \frac{\delta}{\sqrt[3]{r \cdot D_M^2}} + \frac{\mu_T}{4n} \right) \tag{2.14}$$

Where:

- Z_0 is the complex zero-sequence impedance of the conductor in [Ω/ m].
- R_0 is the zero-sequence resistance of the conductor in [Ω/ m].
- X_0 is the zero-sequence inductance of the conductor in [Ω/ m].
- R_1 is the positive sequence resistance of the conductor in [Ω/ m]
- n is the number of sub-conductors per bundle, which is equal to 1
- f is the frequency of the system in [Hz]
- δ is the equivalent soil penetration depth in [m] calculated from Equation (2.13)
- r is the radius of the conductor in [m]
- D_M is the geometrical mean spacing between the conductors in [m]. For trefoil formation a value of the diameter of the insulator is considered. For flat formation, a value of the diameter of the insulator + 70 mm is considered.
- μ_T is the relative permeability of the conductor, which is considered equal to 1 for conductors.

2.4 Losses

Cable losses significantly influence the efficiency and economic viability of a power transmission system. Minimizing these losses enhances energy yield and reduces operational costs.

Although RatedPower does not impose a strict limit on conductor sizing based on losses, a maximum of 5% is generally recommended as a practical upper bound.

Cable losses are categorized into current-dependent and voltage-dependent components. The total active power loss is given by:

$$W_T = 3\,(W_c + W_d) \tag{2.15}$$

Where:

- W_T is the total active power loss of the line [W]
- W_c is the current-dependent loss due to Joule effect [W]
- W_d is the voltage-dependent loss due to the dielectric effect [W]

2.4.1 Current-dependent losses (Joule Effect)

Joule losses occur when current flows through a conductor with resistance, resulting in heat dissipation. The per-phase loss is calculated as:

$$W_c = I^2 R \tag{2.16}$$

Where:

- W_c is the joule effect loss per phase [W]
- I is the operating current [A]
- R is the conductor resistance[Ω]

These losses include:

- Ohmic losses (Joule effect)
- Skin effect losses (included in AC resistance, see Equation (2.6))
- Proximity effect losses (included in AC resistance, see Equation (2.8))
- Metallic sheath losses (neglected in this methodology)

2.4.2 Voltage-dependent losses (Dielectric Losses)

Dielectric losses occur when the alternating electric field in the cable insulation causes small energy dissipation in the form of heat. These losses are influenced by the voltage level, frequency, and the insulating material's properties. *For voltages to earth below 127 kV, RatedPower will neglect the resulting value and only consider current-dependent losses.*

The equation that describes the dielectric losses is the following:

$$W_d = \omega \cdot C \cdot U_0^2 \tan \delta \tag{2.17}$$

Where:

- W_d is the dielectric effect loss [W]
- ω is the angular frequency, $2\pi f$ [rad/s]
- C is the capacitance of the cable given by the manufacturer [F/m]

- U_0 is the operating voltage to earth [V] (Note: this does not refer to the phase to phase nominal voltage)
- $\tan \delta$ is the loss factor of the insulation at power frequency and operating temperature. For XLPE cables with a voltage higher than 36 kV, a value of 0.001 is considered (see IEC 60287-1-1, Table 3 [1])

Chapter 3

Equipment Sizing

In high-voltage underground cable systems, the correct sizing and selection of joints and terminals are essential to ensure the integrity, reliability, and safety of the entire installation. These components serve as critical connection points to link cable sections, while terminals connect cables to equipment such as transformers, switchgear in a substation or overhead lines. Improperly sized or incompatible accessories can lead to overheating, increased electrical losses, or even catastrophic failures.

This section outlines the methodology used to determine the appropriate sizes and types of joints and terminals, taking into account the compliance with the conductor material (aluminum or copper), insulation type (XLPE), current-carrying capacity and thermal performance. The goal is to ensure that all accessories are fully compatible with the cable system and capable of withstanding both normal operating conditions and fault scenarios.

3.1 Terminals

The terminals used in this methodology are from manufacturers that comply with the IEC 60840 standard [6]. These terminals are outdoor, dry-insulated and suitable for substations. An illustration of the terminal's type is shown in Figure 3.1.



Figure 3.1: Outdoor terminal illustration (source: Pfisterer)

In this methodology, terminals are selected by identifying the first available terminal type that meets or exceeds the system's rated voltage, in accordance with standardized voltage ranges, from 46kv up to 550kV. This approach ensures that the terminal is neither under- nor over-specified, optimizing both safety and cost-efficiency.

The terminal selected is the lowest-rated terminal that still fully complies with the required voltage class, ensuring compatibility with the cable insulation and conductor material. This method aligns with IEC standards for high-voltage accessories and guarantees reliable performance throughout the cable system's operational life.

3.2 Joints

The joints are essential components used to connect consecutive cable sections along the route. Their selection must ensure electrical continuity and insulation integrity equivalent to that of the cable itself. In this methodology, joints are selected based on the system voltage and the standardized voltage classes, following the same classification used for terminal selection. This ensures compatibility with the cable insulation and conductor material (aluminum or copper).

The joints used in the design are from manufacturers complying with the IEC 60840 standard [6]. An illustration of the joint is shown in Figure 3.2.



Figure 3.2: Joint illustration (source: Pfisterer)

For each voltage level, the first joint type capable of withstanding the system voltage is selected to ensure both safety and cost-efficiency. Additionally, based on the typical installation practice, one joint is installed every 500 meters. This spacing accounts for practical limitations in cable transport and handling, while maintaining system reliability and minimizing the number of joints, which are potential points of failure.

This approach ensures that all joints are appropriately rated for the electrical and environmental conditions of the installation, contributing to the long-term performance and safety of the HV cable system.

Chapter 4

Civil calculation

The civil aspect of high-voltage underground cable installation are critical to ensuring the long-term safety, accessibility, and performance of the electrical system. One of the most fundamental components of this design is the cable trench, which provides physical protection and structural support for the cables.

This chapter outlines the methodology used to size and design the trench dimensions, taking into account factors such as the number of cables, cable spacing, formation type, and depth of burial. By optimizing the trench layout, the design aims to balance technical performance and construction feasibility.

4.1 Trench sizing

In this solution, the considered installation type is the directly buried cables in a trench. In this case, the trench size calculated for the HV underground cable is determined by three dimensions:

Vertical depth:

The vertical depth of the trench depends directly on the user inputs. The range of depth will be minimum 0.5m and maximum 1.5m.

Horizontal width:

The width of the trench depends on several parameters as follows:

1. **Number of circuits**: The number of circuits is a user input and is ranging from a minimum of 1 and maximum of 3 circuits.
2. **Cable formation**:
    - Trefoil formation: the distance between phase axes is considered as: 2xDe where De is the diameter of the phase in mm. (Or 4x Cables radius).
    - Flat formation: the distance between phases is a fixed input at De + 70mm. Where De is the diameter of the phase conductor in mm.
3. **Distance between circuits or groups of cables**: What is considered as a group is 3 single core cables, regardless of the formation type. The distance between groups or circuits

is measured from the axes as shown in Figure 4.1 and it is a user input ranging from a minimum of 200mm to a maximum of 2000mm.

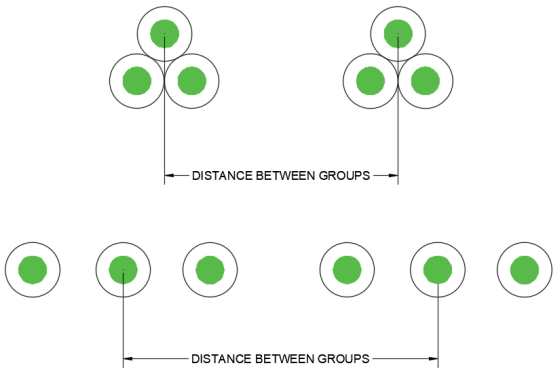


Figure 4.1: Distance between groups

4. **Distance between trench wall and first cable**: A clearance of 50 mm is maintained on both sides of the trench to allow for safe installation and thermal expansion.

The total trench width is calculated by summing the widths of all cables, inter-phase spacings, inter-circuit spacings, and side clearances. The minimum width of a trench is considered to be 0.4m.

Length:

The trench will follow the cables route defined by the user so the length of the trench should be exactly the same as the cables.

The following illustrations are examples of a trench laying for 3 circuits in trefoil 4.2 and a trench for 2 circuits in flat formation 4.3

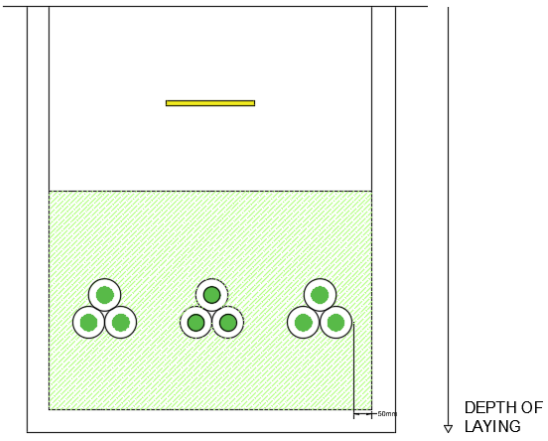


Figure 4.2: Trefoil formation trench example

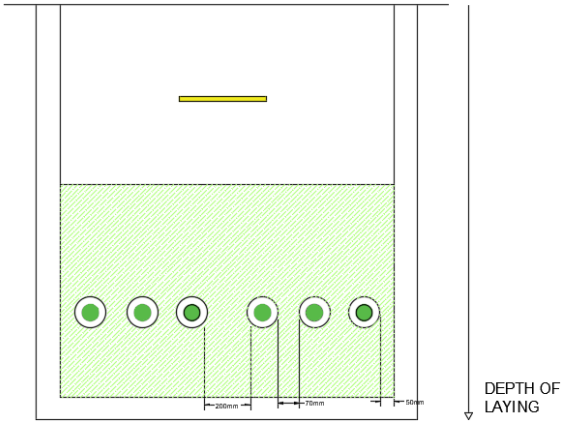


Figure 4.3: Flat formation trench example

This flexible approach allows the trench design to be adapted to various project-specific constraints while maintaining compliance with safety and performance standards.

4.2 Clearances

In this methodology, clearance requirements are only considered in specific crossing scenarios. These include:

Crossing a Switching and Breaking Station Area:

When the underground cable route intersects or passes near a switching and breaking station, a horizontal clearance buffer is maintained between the cable and any station container. This buffer ranges from a minimum of 0.2 meters to a maximum of 0.5 meters. This ensures safe separation of the HV cable for thermal dissipation, maintenance access, and fault protection.

Crossing Under Overhead Lines:

In cases where the underground line crosses beneath an overhead transmission or distribution line, tower spotting is adjusted during the design phase to prevent any tower foundation from intersecting with the underground cable path while keeping a buffer of 0.5m. This proactive coordination avoids conflicts between the two systems and ensures compliance with safety regulations and utility standards.

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Appendix A

Electrical parameters calculation

In this appendix, the process and calculation of the electrical parameters are presented.

The electrical cable used for this example has the following characteristics:

- Number of circuits → 1
- Section → 500 mm^2
- Conductor radius (only conductive material) → 0.0131 m
- Diameter (including insulator) → 0.0444 m
- Insulation → XLPE
- Resistance DC at 20ºC → $6.05 \cdot 10^{-5}\ \Omega/m$
- Inductance → $3.3 \cdot 10^{-7}\ H/m$
- Capacitance → $9.739 \cdot 10^{-8}\ F/m$

A.1 Resistance

The first step is to compute the DC resistance at maximum temperature. For XLPE, 90ºC is the maximum operating temperature. Equation (2.5) is used for this:

$$\begin{gathered} R_{dc\theta} = R_{dc20C}\left(1 + \alpha_c\left(\theta_c - 20\right)\right) = \\ 6.05 \cdot 10^{-5} \cdot \left(1 + 0.00403 \cdot (90 - 20)\right) = 7.757 \cdot 10^{-5}\Omega/m \end{gathered} \tag{A.1}$$

The skin effect is now computed using Equations (2.6) and (2.7)

$$x_s^4 = \left(\frac{8\pi f}{R_{dc\theta}} k_s \cdot 10^{-7}\right)^2 = \left(\frac{8\pi \cdot 50}{7.757} \cdot 1 \cdot 10^{-7}\right)^2 = 2.6246 \tag{A.2}$$

$$y_s = \frac{x_s^4}{192 + 0.8 \cdot x_s^4} = \frac{2.6246}{192 + 0.8 \cdot 2.6246} = 0.01352 \tag{A.3}$$

Similarly, proximity effect is computed from Equations (2.8) and (2.9):

$$x_p^4 = \left(\frac{8\pi f}{R_{dc\theta}} k_p \cdot 10^{-7}\right)^2 = \left(\frac{8\pi \cdot 50}{7.757} 1 \cdot 10^{-7}\right)^2 = 2.6246 \tag{A.4}$$

$$y_p = \frac{x_p^4}{192 + 0.8 \cdot x_p^4}\left(\frac{d_c}{s}\right)^2\left[0.312\left(\frac{d_c}{s}\right)^2 + \frac{1.18}{\frac{x_p^4}{192+0.8\cdot x_p^4} + 0.27}\right] = \frac{2.6246}{192 + 0.8 \cdot 2.6246}\left(\frac{26.2}{26.2}\right)^2\left[0.312\left(\frac{26.2}{26.2}\right)^2 + \frac{1.18}{\frac{2.6246}{192+0.8\cdot 2.6246} + 0.27}\right] = 0.0605 \tag{A.5}$$

Finally the AC resistance of the cable can be calculated from Equation (2.4):

$$R_{ac\theta} = R_{dc\theta}\left(1 + y_s + y_p\right) = 7.757 \cdot (1 + 0.01352 + 0.0605) = 8.331 \cdot 10^{-5}\Omega/m \tag{A.6}$$

A.2 Reactance

For the calculation of the reactance, the datasheet inductance is used, and Equation (2.10) is applied.

$$X_L = 2\pi \cdot f \cdot L = 2\pi \cdot 50 \cdot 3.3 \cdot 10^{-7} \approx 1.0367 \cdot 10^{-4}\Omega/m \tag{A.7}$$

A.3 Impedance

Having the resistance and the inductive reactance of the line calculated in the previous sections, the impedance of the entire line can be calculated as follows:

$$\vec{Z} = R + jX_L = 8.331 \cdot 10^{-5} + j1.0367 \cdot 10^{-4} = 1.33 \cdot 10^{-4}\Omega/m \tag{A.8}$$

A.4 Capacitive susceptance

The capacitive susceptance can also be obtained from the manufacturer's capacitance and the following Equation (2.12)

$$B = 2\pi \cdot f \cdot C = 2\pi \cdot 50 \cdot 3.1 \cdot 10^{-10} \approx 9.739 \cdot 10^{-8}\Omega^{-1}m^{-1} \tag{A.9}$$

A.5 Zero sequence impedance

For the zero sequence parameters, first the soil penetration must be calculated according to Equation (2.13).

$$\delta = \frac{1.851}{\sqrt{2\pi f \frac{\mu_0}{\rho}}} = \frac{1.851}{\sqrt{2\pi \cdot 50 \frac{4\pi \cdot 10^{-7}}{300}}} = 1613.57m \tag{A.10}$$

Then, the zero sequence impedance is obtained from Equation (A.11):

$$Z_0 = R_0 + jX_0 = \frac{R_1}{n} + \frac{3}{4}\pi f \mu_0 + jf\mu_0\left(3 \cdot \ln\frac{\delta}{\sqrt[3]{r \cdot D_M^2}} + \frac{\mu_T}{4n}\right) = \frac{8.33 \cdot 10^{-5}}{1} + \frac{3}{4}\pi \cdot 4\pi \cdot 10^{-7} + j50 \cdot 4\pi \cdot 10^{-7}\left(3 \cdot \ln\frac{1613.57}{\sqrt[3]{0.0131 \cdot 0.0444^2}} + \frac{1}{4 \cdot 1}\right) = \left[2.314 \cdot 10^{-4} + j0.00206\right]\Omega/m \tag{A.11}$$

Appendix B

Underground line conductor sizing

In this appendix, the sizing process of a high-voltage underground cable following the IEC standard will be presented.

The characteristics of the conductor to be sized are the following.

- Aluminum conductor
- Trefoil formation
- Crossbonded
- Frequency of 50Hz
- Operating voltage 46kV
- Operating current of 1255A
- Length of 1000m
- Maximum number of circuits 3
- Maximum voltage drop 5%
- Laying depth of 1m
- Ground temperature of 20°C
- Thermal resistivity of 1 $K \cdot m/W$
- Distance between circuit axes of 200mm
- Cross-sections available: $95mm^2$, $120mm^2$, $150mm^2$, $185mm^2$, $240mm^2$, $300mm^2$, $400mm^2$, $500mm^2$, $630mm^2$, $800mm^2$, $1000mm^2$, $1200mm^2$, $1400mm^2$, $1600mm^2$ and $2000mm^2$.

B.1 IEC standard

As mentioned in Chapter 2, the line sizing is done according to the IEC standard 61089. In particular, it will find the conductor with the least volume that complies with the voltage drop, short-circuit current, and current capacity criteria. To do so, it will evaluate candidate conductors with the cross sections listed above and between 1 and 3 circuits.

The voltage drop of the line is defined according to Equation 2.3. Given the conditions, the conductor with the lowest volume that yields a voltage drop below the limit of 5% has a cross section of $95mm^2$ and 1 circuit.

$$Vd_{3ph}(\%) = \frac{\sqrt{3} \cdot I_b \cdot L \cdot [R \cdot cos\theta + X \cdot sin\theta]}{Vd_{3ph} \cdot N} \cdot 100 \approx 1.9\% \tag{B.1}$$

Where:

- $R \approx 0.411\ \Omega/km$
- $X \approx 0.135\ \Omega/km$
- $\theta = 0\ rad$

With regards to the short-circuit current, for a voltage level of 46kV, the value is fixed to 31500A. Considering that the conductor is built in aluminum, the minimum cross section capable of withstanding the short circuit current is 333 mm^2. Consequently, the conductor with 1 circuit will be the smallest one to comply with this criterion.

$$S = \frac{I_{sc} \cdot \sqrt{t}}{K \cdot \sqrt{\ln\left(\frac{\theta_f + \beta}{\theta_i + \beta}\right)}} \approx 333mm^2 \tag{B.2}$$

Where:

- $\beta = 228\ K$
- $K = 148\ As^2/mm^2$

Lastly, in terms of current cable capacity, the smallest feasible conductor has a cross section of $500mm^2$ and 3 circuits. This is determined by Equation 2.1.

$$I_{ccc} = I_s \cdot d_{ld} \cdot d_{gt} \cdot d_{tr} \cdot d_{dc} \approx 440.2A \tag{B.3}$$

Where:

- d_{ld} is the derating factor due to the laying depth. Set to 1.0.
- d_{gt} is the derating factor due to the ground temperature. Set to 1.0.
- d_{tr} is the derating factor due to the thermal resistivity of the ground. Set to 1.0.
- d_{dc} is the derating factor due to the distance between circuit axes. Set to 0.71.
- Is is the ampacity under standard conditions according to IEC. Set to 620A.

In order for the conductor to be capable of withstanding the operating current, 3 circuits are needed. Yielding a final current cable capacity of 1320.6A.

Once calculated the minimum requirements to withstand all criteria, the most restrictive one defines the final configuration, thus the final size of the conductor will be $500mm^2$ and 3 circuits.

Appendix C

Underground line losses

In this appendix, the electrical losses calculation is presented. The characteristics of the conductor for which the electrical losses will be calculated are:

- AC resistance $\longrightarrow 8.331 \cdot 10^{-5}\ \Omega/m$
- Capacitance $\longrightarrow 1.7 \cdot 10^{-10}\ F/m$
- Length $\longrightarrow 40000\ m$
- Voltage $\longrightarrow 200\ kV$
- Operating current $\longrightarrow 1255\ A$
- Total capacity $\longrightarrow 434.745 MVA$

The current-dependent losses due to Joule effect are calculated following Equation (2.16):

$$W_c = I^2 R = 1255^2 \cdot 8.331 \cdot 10^{-5} \cdot 40000 = 5248613.3W = 5.2486MW \tag{C.1}$$

The voltage-dependent losses due to dielectric effect are computed according to Equation (2.17):

$$W_d = \omega \cdot C \cdot U_0^2 \tan\delta = 2\pi \cdot 50 \cdot 1.7 \cdot 10^{-10} \cdot 40000 \cdot \left(\frac{200000}{\sqrt{3}}\right)^2 \cdot 0.001 = 28483.77W = 0.0285MW \tag{C.2}$$

The total loss is therefore calculated as the sum of the previously calculated losses:

$$W_T = W_c + W_d = 5.2486 + 0.0285 = 5.2771MW \tag{C.3}$$

This result can also be presented as a percentage of the total capacity of the line:

$$W_T(\%) = \frac{W_T}{S} \cdot 100 = \frac{5.2771}{434.745} \cdot 100 = 1.214\% \tag{C.4}$$

Appendix D

Trench sizing

In this appendix, the sizing process of a trench for the underground line is presented. For the sake of illustrating the calculation, a line with the following configuration has been considered:

- 3 circuits
- 800mm distance between circuit's axes
- Flat formation
- 108mm conductor diameter with insulation

For a cable in flat formation with the aforementioned configuration, the width of a single circuit is computed as follows:

$$w_c = 3 \cdot D_e + 140 = 468mm \tag{D.1}$$

Where:

- D_e is the diameter of the conductor with insulation

Since we have more than one circuit, we compute the distance between the circuit edges, which corresponds to the distance between the edges of two adjacent circuits:

$$d_{ce} = d_{ca} - w_c = 336mm \tag{D.2}$$

Where:

- d_{ca} is the distance between circuit's axes

The conductor considered had 3 circuits, thus the width required to fit it is defined as:

$$w_t = (N - 1) \cdot d_{ce} + N \cdot w_c + 100 = 2164mm \tag{D.3}$$

Where:

- N is the number of circuits.

It is worth noting that an extra 100mm are added to ensure 50mm between the first and last circuit and the walls, leading to a resulting trench of 2.164m width.